AARAV SHARMA

Kochi, Kerala | aarav.sharma.dev@gmail.com | +91 98765 43210 | github.com/aarav-ai

PROFESSIONAL SUMMARY

Proactive Python AI & Data Science enthusiast with hands-on experience in building LLM-powered document intelligence engines, RESTful APIs, and retrieval-augmented generation (RAG) pipelines. Proficient in Python 3.11, pypdf, python-docx, scikit-learn, and prompt engineering. Strong foundation in linear algebra, algorithms, and modular OOP architecture.

TECHNICAL SKILLS

- **Languages & Core:** Python 3.10+, SQL, Bash, C++ (Foundations)
- **Libraries & Data:** Pandas, NumPy, Scikit-Learn, Regular Expressions (re), pypdf, python-docx, Matplotlib
- **GenAI & LLM Tooling:** OpenRouter API, OpenAI API, HuggingFace, ChromaDB, FAISS, Prompt Engineering, Pydantic Schema Validation
- **Web & Developer Tools:** FastAPI, Requests, Git/GitHub, Docker, Linux/Unix CLI, Virtual Environments (venv), Pylint

EXPERIENCE & INTERNSHIPS

Applied AI Engineering Intern - CogniTech Labs, Kochi *Jan 2024 - Jun 2024 (6 Months)*

- **Document Parsing Engine:** Designed modular Python ingestion pipeline extracting tabular and multi-column text from 15,000+ PDFs/DOCX using pypdf and python-docx, achieving 98.4% extraction accuracy.
- **LLM Pipeline Optimization:** Engineered token-efficient prompt templates for OpenRouter and OpenAI models with exponential backoff retries, slashing per-request token consumption by 34%.
- **FastAPI Microservice:** Integrated the evaluation workflow into high-throughput FastAPI endpoints validated strictly via Pydantic schemas, reducing API response latency to under 120ms.

PROJECTS & PRACTICAL IMPLEMENTATION

DocuIntellect - Multi-Format Document Intelligence Pipeline *Python, pypdf, ChromaDB, FastAPI*

- **End-to-End System:** Engineered complete CLI & API pipeline to ingest, clean, chunk, and embed contracts and technical specifications. Implemented cosine similarity semantic filtering with scikit-learn.
- **Robustness & Code Hygiene:** Wrote 40+ unit tests with pytest (88% code coverage) and maintained 9.8/10 Pylint score across the codebase adhering to strict PEP 8 standards.

Semantic Code Search Engine (RAG) *Python, OpenRouter, FAISS, Streamlit*

- **Hybrid Search:** Combined sparse BM25 keyword matching with dense FAISS vector embeddings to query 50+ open-source Python repositories in real-time.

ACADEMIC BACKGROUND & FOUNDATIONS

B.Tech in Computer Science & Engineering (Data Science Specialization) *2021 - 2025 (Graduating May 2025)*

Kerala Technological University (KTU) | CGPA: 8.85 / 10.0

- **Core Coursework:** Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Natural Language Processing, Deep Learning.